

C1GALT1C1 (COSMC) — Function-Assignment Hypothesis Report

Gene: C1GALT1C1 (COSMC) · **UniProt:** Q96EU7 · **Organism:** Homo sapiens (NCBITaxon:9606)
Term under review: GO:0016263 — N-acetylgalactosaminide beta-1,3-galactosyltransferase activity (MF) **Existing annotation:** IMP,

P 37216524

negated = true Seed hypothesis: *C1GALT1C1 does not have N-acetylgalactosaminide beta-1,3-galactosyltransferase activity (GO:0016263).*

Executive Judgment

Verdict: SUPPORTED (high confidence). C1GALT1C1 encodes **Cosmc**, a **private ER molecular chaperone** that is required for the folding/stability of the core 1 β 3-galactosyltransferase (**C1GALT1 / T-synthase**, UniProt Q9NS00). The catalytic activity GO:0016263 (synthesis of Gal β 1-3GalNAc- α -Ser/Thr, the T antigen) belongs to **C1GALT1, the client enzyme — not to C1GALT1C1**. Multiple lines of primary evidence, and an explicit UniProt CAUTION, show that C1GALT1C1 **has no transferase activity**. The negated annotation is therefore **justified and should be retained**.

Key caveat on evidence code: the annotation uses **IMP** with negation. The mutant phenotype in the reference

(
P 37216524

) shows *loss of T-synthase (C1GALT1) activity* when Cosmc is defective. That result demonstrates Cosmc's **chaperone requirement** for the reaction; it does not show Cosmc itself catalyzes it. IMP is defensible for a NOT-annotation here, but the *positive* function of this gene is chaperone activity, and that is what should carry the informative MF term.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Con- lim
UniProt Q96EU7 (CAUTION)	Database/ curation	Supports	Does C1GALT1C1 have GT activity?	"Was originally assigned to be a glycosyltransferase. However, it was later shown that it has no transferase activity and rather acts as a chaperone. " KW=Chaperone (no Transferase).	Human, reviewed entry	Hig dat but pri
Ju & Cummings 2002, P 12464682	Direct assay / mutant complementation	Supports	Is the transferase activity intrinsic to Cosmc?	Cosmc restores C1 β 3Gal-T activity in Cosmc-deficient Jurkat & insect cells; C1GALT1 gene/mRNA is normal in those cells; Cosmc "acts as a specific molecular chaperone in assisting the folding/ stability of C1 β 3Gal- T." The transferase is a separate gene product.	Human Jurkat, Hi-5 insect cells	Hig fou fun stu
Aryal, Ju & Cummings 2010, P 19923218	Direct in vitro assay	Supports	Does Cosmc perform folding vs catalysis?	Cosmc directly refolds denatured T- synthase in vitro (ATP-independent, substrate-specific); "first ER chaperone required for folding of a glycosyltransferase." Tn-syndrome Cosmc	Recombinant proteins in vitro	Hig cha act

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Con- lim
Aryal, Ju & Cummings 2014, P 24616093	Interaction / mechanism	Supports	Mechanism of Cosmc– client recognition	mutant has reduced chaperone function. Cosmc recognizes a specific stem-region motif (CBRT) in T- synthase; grafting CBRT onto β 4GalT1 confers Cosmc binding — a chaperone recognition system, not catalysis.	Human proteins, in vitro	Hig
Erger et al. 2023, P 37216524 (the IMP reference)	Mutant phenotype (germline)	Supports	What does a germline C1GALT1C1 variant cause?	Titled " Germline C1GALT1C1 mutation causes a multisystem chaperonopathy "; A20D-Cosmc reduces C1GALT1 (T- synthase) activity and impairs O- glycosylation → aHUS-like disease. Loss of chaperone function, not loss of intrinsic transferase activity.	Human patients (2 half- brothers)	Hig rol cha
Crew et al. 2008, P 18537974	Mutant phenotype (human)	Supports/ qualifies	Consequence of inactivating C1GALT1C1 variants	Inactivating variants (E152K, S193P, M1I) produce Tn phenotype via T- synthase deficiency — i.e., Cosmc loss disables the transferase,	Human erythroblasts	Hig

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Con- lim
				consistent with a chaperone role.		
This report — pairwise alignment	Computational (evolutionary)	Qualifies	Why was GT activity ever attributed?	COSMC (318 aa) vs C1GALT1 (363 aa) ≈ 33% identity ; both in GT31 (β3-Gal-T subfamily). Explains original name "C1Gal-T2" and the GT misannotation (PubMed:12361956).	Sequences Q96EU7 / Q9NS00	Me coa alig con Un ass
QuickGO API (this report)	Database provenance	Supports	Where does GO:0016263 actually sit?	C1GALT1C1/Q96EU7: single GO:0016263 = NOT enables , IMP, P 37216524 (the annotation under review). C1GALT1/Q9NS00: GO:0016263 = positive enables via IDA (P 11677243) + IBA + IEA.	EBI QuickGO, human	Hig ter exp gro the neg cha
Ju et al. 2002, P 11677243	Direct assay (IDA source)	Supports	Which gene product is the enzyme?	"Cloning and expression of human core 1 β1,3-galactosyltransferase" — establishes C1GALT1 as the GO:0016263 catalyst.	Human, recombinant enzyme	Hig

GO Curation Implications (leads — require curator verification)

- **Retain the negated GO:0016263 (MF) on C1GALT1C1.** The evidence robustly supports that this gene product lacks N-acetylgalactosaminide β -1,3-galactosyltransferase activity. The negation prevents paralog carry-over of the T-synthase activity onto its chaperone.
- **Positive MF that IS supported:** chaperone activity — candidate terms **GO:0051082 (unfolded protein binding)** and/or **GO:0044183 / GO:0140662-type** chaperone terms; most specifically the role assisting C1GALT1 folding. Direct evidence:

P 19923218

P 12464682

Avoid "protein binding" as the final term — "unfolded protein binding / chaperone" is more informative.

- **Supported BP:** protein folding (GO:0006457) and **O-glycan / T-antigen biosynthesis via enabling C1GALT1** (e.g., GO:0016266 O-glycan processing) as an *involved-in / contributes-to* relationship, not as a direct catalytic step.
 - **Supported CC:** endoplasmic reticulum membrane; single-pass type II membrane protein (GO:0005789).
 - **Evidence-code note (lead):** consider whether the NOT-GO:0016263 is better supported as **IC/ND** or kept as **IMP**; the mutant data are about the *client's* activity, so IMP for a NOT-MF is acceptable but worth a curator note.
-

GO Decision Table (lead — curator to verify)

GO term	Aspect	Gene	Current / proposed	Recommended action	Evidence
GO:0016263 N-acetylgalactosaminide β -1,3-galactosyltransferase activity	MF	C1GALT1C1	NOT enables, IMP, P 37216524	RETAIN negation	No intrinsic transferase activity; UniProt CAUTION; P 12464682 / 19923218
GO:0016263 (same)	MF	C1GALT1 (paralog Q9NS00)	enables, IDA P 11677243 (+IBA/IEA)	Keep (belongs here) — reference point	Direct assay of the actual enzyme
GO:0051082 unfolded protein binding	MF	C1GALT1C1	(candidate to add/confirm)	ADD / confirm as informative positive MF	P 19923218 , P 12464682 , P 24616093
GO:0006457 protein folding	BP	C1GALT1C1	(candidate)	ADD/confirm — direct chaperone role	P 19923218
GO:0016266 O-glycan processing (or 'contributes to' C1GALT1 activity)	BP	C1GALT1C1	(candidate)	ADD as <i>involved-in/ contributes-to</i> , not direct catalysis	P 37216524 , P 24643043
GO:0005789 endoplasmic reticulum membrane	CC	C1GALT1C1	(candidate)	Confirm; single-pass type II membrane	UniProt Q96EU7

Bottom line: retain the negated GO:0016263 on C1GALT1C1; the gene's informative *positive* molecular function is chaperone / unfolded-protein binding, not any transferase term.

Mechanistic Scope

- **Immediate molecular function of C1GALT1C1:** ATP-independent, client-specific **molecular chaperone** in the ER that binds a stem-region motif (CBRT) of C1GALT1 and promotes its folding/stability, preventing proteasomal degradation.
- **Not the gene product's activity:** the galactosyl transfer reaction itself (Gal β 1-3GalNAc-Ser/Thr synthesis) — that is enzymatically executed by **C1GALT1**.
- **Downstream / indirect (not to be conflated with MF):** T-antigen formation, extended O-glycan biosynthesis, Tn/STn antigen exposure on Cosmc loss, and disease phenotypes (Tn polyagglutination syndrome, aHUS-like chaperonopathy). These arise from *loss of chaperone function*, not from loss of an intrinsic transferase.

Conflicts and Alternatives

- **Historical database carry-over (resolved):** the original 2002 report (PubMed:12361956) named the protein "Core 1 β 1,3-galactosyltransferase 2 (C1GalT2)" based on GT31 paralogy (~33% identity to C1GALT1). This is the source of any residual GT annotation; it was explicitly refuted (UniProt CAUTION;

P 12464682

No credible primary evidence supports intrinsic transferase activity.

- **Paralog confusion (the main risk):** C1GALT1 (Q9NS00) is the true GO:0016263 enzyme. Annotation pipelines keyed on GT31 family membership could re-propagate the term; the explicit negation guards against this.
- **No organism-specific or isoform-specific evidence** contradicts the chaperone-only interpretation; the chaperone role is conserved across vertebrates (client-specificity data,

P 24616093

Knowledge Gaps

1. **Formal enzymatic exclusion assay.** Checked: no primary paper reports a direct in vitro assay of purified Cosmc acting as a galactosyltransferase; activity has only ever been observed via C1GALT1. Matters because the negation rests on *absence of evidence* + *mechanistic understanding*. Resolved by an in vitro UDP-Gal → GalNAcα-O-phenyl assay with purified Cosmc alone (expected: no activity).
 2. **Structural confirmation of a degenerate catalytic site.** Checked: UniProt lists GT31 similarity but no experimental structure with catalytic-residue analysis surfaced in searches. A structure/model confirming loss of the catalytic/acceptor-binding residues would strengthen the mechanistic basis.
 3. **Evidence-code appropriateness for the NOT annotation.** Whether IMP vs IC/ND best fits a negative MF; a curator judgment, not an experimental gap.
-

Discriminating Tests

- **Direct transferase assay of purified Cosmc** (UDP-[3H]Gal donor, GalNAcα-O-phenyl acceptor) with C1GALT1 as positive control — cleanly separates "no intrinsic activity" from "chaperone only."
 - **Reciprocal reconstitution:** active T-synthase produced only when Cosmc is co-expressed in Cosmc-null (insect) cells — already shown
([P 12464682](#));
a targeted re-run isolates the chaperone dependency.
 - **Structure/AlphaFold catalytic-residue mapping:** compare Cosmc vs C1GALT1 active-site and donor-binding residues to show catalytic degeneration.
-

Curation Leads (verify before applying)

- **Action:** Keep negated GO:0016263 (MF) on C1GALT1C1. **Do not** add any positive galactosyltransferase term.

- **Add positive MF:** unfolded protein binding / chaperone activity (e.g., GO:0051082) — refs

P 19923218

P 12464682

- **Add/confirm BP:** protein folding (GO:0006457); contributes-to O-glycan biosynthesis (via C1GALT1).
- **Add/confirm CC:** endoplasmic reticulum membrane (GO:0005789), integral/single-pass type II.
- **Candidate references w/ exact snippets to verify:**

P 12464682

— "Cosmc acts as a specific molecular chaperone in assisting the folding/stability of C1β3Gal-T."

P 19923218

— "Cosmc represents the first ER chaperone identified to be required for folding of a glycosyltransferase."

P 37216524

— title "Germline C1GALT1C1 mutation causes a multisystem chaperonopathy."

- UniProt Q96EU7 CAUTION — "...it has no transferase activity and rather acts as a chaperone."
- **Suggested curator question:** Is IMP the preferred evidence code for this NOT-MF, or should IC/ND be used given the phenotype reflects the client enzyme's activity?
- **Suggested experiment:** direct in vitro transferase assay of purified Cosmc (expected negative).

Provenance

- Literature via PubMed (PMIDs above); entry data via UniProt Q96EU7 (.txt) and Q9NS00 (.fasta).

- Computed pairwise global alignment COSMC vs C1GALT1: ~**33% identity** (318 vs 363 aa), GT31 paralogs — executed in this run (Needleman–Wunsch over fetched UniProt sequences).

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery